

- as inputs two integer indices (i,j) and returns *True* if (S[i]...S[j]) constitutes a valid English word. Else it returns *False*. Can you write a function that segments the sequence S to obtain the original sentence? What is the time and space complexity of your approach?
4. Given a set A of N integers, and an integer value K, you are asked to find out whether there is a subset S of A which exactly sums to the value K. What is the time and space complexity of your algorithm? Instead of finding a subset S which sums to K exactly, can you identify whether any subset S exists whose sum is greater than or equal to K? Can you leverage your earlier solution for this problem?
 5. We are all familiar with vector spaced representations of words in natural language processing. Popular pretrained word embeddings are word2vec and glove embeddings. What are the advantages and disadvantages of word embeddings vs sub-word/character embeddings? If you are told that your model can be expected to encounter new text not seen in training data, which will you prefer?
 6. Following up on the question above, you are asked to generate sub-word embeddings for a given corpus of text. Can you outline the algorithm you will use for the same? Assume that you can use the word frequency count to decide which sub-word embeddings to choose for encoding. Two popular sub-word embeddings algorithms are byte pair encoding and word piece encoding techniques. Byte pair encoding is described in the paper 'Neural Machine Translation of Rare Words with Subword Units'. Word piece encoding is described in the paper 'Japanese and Korea Voice Search'. It will be useful to understand the differences between these two techniques as well as understand how to use each. Many NLP models, including BERT and its variants, use byte pair encoding or similar approaches for the representation of embeddings.
 7. Named entity recognition is a well-known problem in NLP. Given a piece of text, you need to identify different types of entities such as PERSON, LOCATION, COMPANY, PRODUCT, etc. Both rule based and machine learning techniques have been proposed to solve the problem. This is an example of a sequence tagging problem, where we are given a sentence/piece of text consisting of word tokens, and the task is to tag it using a specified set of tags. Can you come up with a simple approach to solve the sequence tagging problem? Popular ML approaches for sequence tagging include Hidden Markov Models. Here, the idea is to depend on the history of the words we have seen so far and the current word, to decide what could be the tag associated with this word. Conditional random fields are another approach, where we use the information of the set of previous tags generated in the sequence along with the current word, to decide what should be the tag for the current word. What are the advantages and disadvantages of each of these approaches?
 8. Deep learning based approaches have also been used for Named Entity Recognition. Neural NER techniques typically use recurrent neural networks such as LSTMs for sequence tagging. Can you combine LSTM with conditional random fields to provide a hybrid approach for neural Named Entity Recognition?
 9. Support Vector Machines (SVMs) are one of the fundamental machine learning based classifiers. However, they are not probabilistic classifiers. Hence, given a binary classification problem, you can use a SVM to get a prediction on the target class, but you can't get the probabilities associated with each of the target classes. How do we generate probabilities for target classes during prediction from a non-probabilistic classifier such as SVM?
 10. Instead of using a single classifier to make prediction decisions, one approach is to use an ensemble of classifiers. Bagging and boosting are two important techniques in ensemble based classification. Can you explain the differences between the two? Give an example classifier for each of these two techniques. Can you combine these two techniques into a single ensemble?
 11. For classical machine learning models, feature engineering (extracting features from data) is a cumbersome task. If you are given a text classification problem (given a piece of text, you need to associate a category with that text among K different categories), what is the simplest feature representation you can use for this problem? What are the advantages and disadvantages of your approach? Instead of classification, let us assume that you are asked to solve a complicated problem such as reading comprehension. Given a question and a paragraph, you need to find the answer span in the given paragraph. How will you change your feature representation of the input text for this problem?
 12. For the text classification problem, you are given the choice of using a Convolutional Neural Network (CNN), Recurrent Neural Network (RNN) and a transformer based approach. What will you choose, and why?
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